AI ROI LEDGER

Know whether each AI deployment is paying for itself.

roi-ledger-pi.vercel.app

WHAT IT IS

AI ROI Ledger tracks every AI deployment a company runs (agents, automation workflows, AI tools) and answers one question for each: **is it paying for itself?** It compares what the deployment costs (build, hosting, AI tokens and the staff time still spent checking it) with the value it returns against how the work was done *before* AI. It then shows net return, ROI, payback date, cost per run and a clear **Scale / Fix / Kill / Watch** flag. Every figure opens up to show its inputs and formula, so it can be defended in a budget meeting.

WHO IT'S FOR

Operations and finance leads who need to know which AI projects to expand and which to stop.
AI and automation owners proving the value of what they built.
Consultants measuring results for clients after go-live.
Builders who want runs and costs reported automatically from their tools.

HOW IT WORKS

1. Sign in

One click. Set your organisation's name, currency (GBP, EUR, USD) and staff hourly cost.

2. Register

Add a deployment: what it does, the task it replaced, its owner and go-live date.

3. Set the baseline

How the task was done before AI: volume, minutes per task, hourly cost, errors.

4. Feed it data

Runs, costs and human review time, from connectors, the API, a CSV or quick forms.

5. Read the verdict

Cost, value, net, ROI, payback and a flag, refreshed daily, every number explained.

6. Act on it

Alerts when a flag changes or cost per run spikes, plus a monthly summary.

THE NUMBERS IT GIVES YOU

FIGURE	HOW IT IS WORKED OUT	WHAT IT TELLS YOU
Cost to date	One-off costs (build, setup, training) + running costs (AI tokens, hosting, licences, spread by day) + human time (review, fixes, escalations × hourly rate).	The full cost, including the people still needed around the AI.
Value to date	Completed runs × baseline minutes per task ÷ 60 × hourly rate, plus errors avoided × cost per error when both error rates are known.	Staff time and rework the deployment has saved.
Net and ROI	Net = value − cost. ROI = net ÷ cost.	Whether it is ahead, and by how much per £1 spent.
Payback date	First day cumulative value passes cumulative cost; otherwise projected from the last 30 days.	When the investment is (or will be) recovered.
Cost per completed run	Total cost ÷ runs that got the work done (including those a person finished).	Unit cost to compare across deployments and over time.
Flag	Kill → Fix → Scale → Watch, using thresholds you can change.	What to do next, with the reasons listed.
Confidence	Medium when the baseline was measured and runs cover 80%+ of days; otherwise Low.	How far to trust the result before acting on it.

GETTING DATA IN: WORKS WITH ANY STACK

Quick forms. Costs and a weekly human-time entry, typed in under a minute.

CSV import. Templates for runs, costs and time, with a checked preview before anything is saved.

Push API. One call at the end of each run from any code, Make, Zapier or Power Automate. Re-sending never duplicates.

Langfuse. Pulls AI cost, tokens and model per run every day.

OpenRouter. Pulls daily AI spend per API key.

n8n. Pulls workflow executions; ready-made templates report runs, collect weekly time and send alerts.

CAPABILITIES

Every number explained. Click any figure to see its formula, inputs and any assumptions made.

Counted once. Human review time is a cost; value is the full time saved. Nothing is deducted twice.

Daily snapshots. Data syncs and results recalculate every night, building a history over time.

Alerts out. Signed webhooks for flag changes, cost-per-run spikes and the monthly summary.

Secure by design. Each organisation sees only its own data; keys are encrypted and never shown again.

Metadata only. Prompts and AI outputs are never stored: just counts, timings, status and cost.

EXAMPLE (ILLUSTRATIVE FIGURES)

Procurement assistant. Requests used to take days by email: about 600 a month at 30 minutes of staff time each, at £40 an hour, so £20 of staff time per request. With the assistant, around 570 a month complete (about £11,400 of value). It cost £25,000 to build, about £600 a month to run, and staff spend roughly 20 hours a month reviewing it (£800). **Result:** payback in about 2.5 months; after six months, about £35,000 net, ROI around 105%, flag **Scale**. Confidence stays **Low** until the baseline is measured rather than reconstructed: the ledger says so, and shows why.

AI ROI LEDGER

Getting started: a step-by-step guide for anyone using the app.

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Allow about 20 minutes for your first deployment. You can come back to any step later; results update as soon as you save. Rough numbers are fine to start with: the app marks them as estimates and shows lower confidence.

#	STEP	WHERE	WHAT TO DO
1	**Sign in and set up**	Go to **roi-ledger-pi.vercel.app** → **Sign in**.	First time only: enter your organisation's name, choose its currency (**GBP, EUR or USD**) and the **default loaded hourly rate** (salary plus about 30% overheads, e.g. £40). Changing currency later does not convert amounts.
2	**Register a deployment**	Deployments → **Register deployment**	Name it, choose the type (agent, workflow, seat tool, custom GPT), set status **Live** and the **go-live date**. Describe the task it replaced and add the owner's email. Under *Data source mapping*, add the IDs your tools use (n8n workflow ID, Langfuse tag, OpenRouter key label) so data finds it.
3	**Add the baseline**	Deployment → **Baseline**	How the work was done before AI: tasks per month, hands-on minutes per task, hourly rate, and error rates if known. No records? Count requests in the old inbox or queue for 2–3 months and ask 3–5 people for typical minutes; choose **Reconstructed**. The page includes guidance.
4	**Record costs**	Deployment → **Log time / cost**	Add **one-off** costs (build, setup, training) with their date, and **recurring** costs (hosting, licences, AI tokens) with the period they cover. Recurring costs are spread evenly across their days.
5	**Log human time weekly**	Deployment → **Log time / cost**	Minutes your team spent last week reviewing output, fixing mistakes, handling escalations and maintaining it. This is the honest cost of the AI; it takes under a minute. (The n8n check-in template can email owners for it.)
6	**Bring in runs**	**Settings** or **Import**	Pick one: a **connector** (Settings → Connectors → add n8n, Langfuse or OpenRouter, then **Sync now**); the **push API** (Settings → Ingest keys → create a key for your developer or automation); or a **CSV** (Import → download template → upload → check preview → import). No run data? Value is estimated from the baseline.
7	**Read the dashboard**	Deployments → your deployment	Six headline figures, a cost-versus-value chart with the payback point, cost split by layer, run statistics and the baseline. Click **Explain** under any number to see exactly how it was calculated. **Recalculate** refreshes it now; otherwise it updates every night.
8	**Stay informed**	**Settings**	Adjust the flag thresholds to your appetite for risk. Add a **webhook** (for example the n8n alerts template) to be told when a flag changes, cost per run jumps, or the monthly summary is ready.

READING THE FLAG

FLAG	WHEN	WHAT TO DO
Scale	ROI above 100% and value at or above the business case.	Roll it out further.
Fix	People touch more than 20% of runs, or it loses money despite normal use.	Improve prompts, rules or process.
Kill	Live 90+ days, losing money, and used at under 30% of the old volume.	Stop it or rethink it.
Watch	None of the above yet.	Keep collecting data.

READING THE CONFIDENCE

CONFIDENCE	MEANING
Medium	Baseline was measured, and run data covers at least 80% of days since go-live.
Low	Baseline reconstructed or estimated, or gaps in run data. Results are indicative.
High	Reserved for controlled comparisons (e.g. staggered roll-out). Coming later.

GOOD TO KNOW

Nothing is counted twice. Review and fix time is always a cost; value is always the full time the baseline says the task took.

Keys stay secret. Connector keys are encrypted and never shown again; ingest keys are shown once, so copy them when created.

Re-sending is safe. Runs, costs and CSV rows with the same ID update the existing record instead of adding a new one.

Pilot access. During the pilot, everyone signs in to one shared workspace. Email sign-in can be switched back on at any time.